

EAPO-NET: Adaptive Energy-Aware Profit Optimization in Next-Generation RANs

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Abstract—The global deployment of more than 3.5 million 5G base stations has made the energy efficiency of Radio Access Networks (RANs) a pressing challenge for mobile network operators (MNOs). Achieving sustainable operation requires jointly optimizing energy consumption, service revenue, and QoS compliance under highly dynamic traffic demands and heterogeneous service requirements.

In this paper, we present EAPO-NET, a Multi-Agent Deep Reinforcement Learning (MADRL) framework that optimizes energy consumption and network profit across the O-RAN control hierarchy. The key insight behind EAPO-NET is a heterogeneous multi-agent decomposition that aligns learning agents with distinct O-RAN control functions. Specifically, EAPO-NET jointly trains RU xApp agents, admission-control xApp agent, and node-level dApp agents using centralized training with decentralized execution (CTDE) and Value Decomposition Networks (VDN), while enabling fully decentralized inference without inter-agent communication. This design preserves scalability and deployment practicality in large-scale O-RAN environments.

Extensive simulations demonstrate that EAPO-NET learns near-optimal control policies that simultaneously reduce energy consumption, maximize network profit, and satisfy QoS constraints. Compared with exact integer linear programming (ILP) formulations, EAPO-NET achieves comparable optimization quality while substantially improving scalability and adaptability to time-varying network conditions.

I. INTRODUCTION

The rapid growth of mobile data traffic and heterogeneous service demands has accelerated the large-scale deployment of 5G networks, with more than 3.5 million base stations (BSs) already in operation worldwide [1]. While this densification improves capacity and coverage, it also leads to a significant increase in energy consumption. In particular, Radio Access Network (RAN) infrastructures account for the majority of network power usage, with Radio Units (RUs) contributing up to 80% due to their continuous involvement in physical layer (PHY) operations such as transmission, reception, and signal processing [1], [2]. As 5G evolves toward wider bandwidths and denser deployments, improving the energy efficiency of RUs becomes a critical challenge. The disaggregated and virtualized nature of the O-RAN architecture enhances the flexibility and programmability of RAN resource management [3], [4]. By separating hardware and software functions and introducing

intelligent control entities such as the Near-RT RIC and O-Cloud, O-RAN enables dynamic orchestration of radio and computing resources [4]. Although O-RAN was not specifically designed for energy efficiency, its architectural flexibility enables the implementation of advanced optimisation strategies, including energy-aware RU operation and adaptive DU/CU placement. These mechanisms are particularly relevant in networks supporting heterogeneous services with varying priority levels and distinct QoS requirements [5].

Empirical studies of operational 5G networks show that traffic is inherently bursty, with alternating periods of high activity and short idle intervals at the millisecond timescale [6], [7]. This fine-grained variability highlights the need for fast, adaptive control mechanisms capable of reacting to rapid traffic fluctuations, for instance by exploiting short transmission gaps for micro-level energy savings [8], [9]. At a coarser timescale, traffic demand also varies significantly over seconds to minutes due to user mobility and diurnal patterns. These longer-term variations enable more structural energy-saving strategies. For example, RUs serving only a small number of user equipments (UEs) during off-peak periods can be switched to sleep mode¹, with their traffic steered to neighbouring RUs when coverage allows it [10]. Similarly, lightly loaded capacity-layer RUs can be temporarily deactivated by handing over their UEs to macro cells [11]. These examples highlight the need for intelligent, context-aware decision-making strategies that jointly consider traffic conditions, user distribution, and service requirements.

In practice, RU sleep mode is tightly coupled with resource allocation and Virtual Network Function (VNF) placement at the Distributed Unit (DU) and Central Unit (CU), which are deployed on the O-Cloud [3], [4]. Joint optimization of these functions enables not only increased RU sleep opportunities but also improved O-Cloud resource consolidation, thereby reducing overall energy consumption. At the same time, such optimization must ensure that QoS requirements across different

¹In this work, *sleeping mode* refers to switching OFF the entire BS/RU, i.e., placing the RU in an OFF state where its main hardware components are powered down. This should not be confused with *blanking*, which refers to deliberately disabling or muting a portion of available RBs to save energy while the BS remains active.

service types are satisfied, while accounting for the economic impact on network operators [5]. The O-RAN architecture enables this through xApps deployed in the Near-Real-Time RAN Intelligent Controller (Near-RT RIC) for near-real-time radio resource control, and dApps acting directly on the O-Cloud for VNF placement decisions [4], [12].

In this paper, we propose a joint optimization framework for RU sleep mode, UE admittance control, and DU/CU² VNF placement in a 5G O-RAN system. The objective is to maximize a net profit function defined as the revenue collected from admitted UEs whose downlink throughput demands are satisfied, in accordance with their SLA contracts, net of the operational energy cost incurred by active RUs, nodes, and instantiated VNFs. This formulation captures the fundamental trade-off faced by the MNO — admitting more UEs increases revenue but activates additional infrastructure, while aggressively sleeping RUs reduces energy cost at the risk of rejecting UEs and forfeiting their associated SLA revenue — while satisfying the QoS requirements of heterogeneous UEs. To attain this objective, we propose EAPO-NET, a novel MADRL-based framework that natively maps onto the O-RAN architecture, instantiating Near-RT RIC xApps for radio resources, a Near-RT RIC xApp for admittance UE control, and O-Cloud dApps for node activation and DU/CU VNF placement.

EAPO-NET employs a centralized training with decentralized execution (CTDE) architecture from Value Decomposition Networks (VDN) [15] built upon the model-free DQN algorithm. Taking as input a representation of current network conditions, EAPO-NET includes three types of cooperative agents — RU xApp agents, an admission xApp agent, and nodes dApp agents — each maintaining a local policy network that observes only its own domain-specific state and selects actions independently at execution time. During training, a centralized critic jointly processes the observations and actions of three types of agents to compute a shared loss, enabling coordination without requiring inter-agent communication at runtime. This decoupling allows EAPO-NET to scale to real O-RAN deployments while retaining the global optimization perspective needed to balance RU energy efficiency and O-Cloud consolidation against UE QoS demands.

A key challenge in designing EAPO-NET is effectively modeling the interactions among RU xApp agents, the admission xApp agent, and the nodes dApp agents. In our O-RAN system, radio-level energy savings are achieved through RB blanking and RU sleep mode, while compute-level resource consumption is managed by DU/CU placement and O-Cloud node activation de-

cisions. Additionally, the admission agent focuses on maximizing MNO profits. Training these agents in isolation can lead to suboptimal coordination; for instance, putting an RU to sleep eliminates its RB blanking budget, while spinning up an additional O-Cloud node to host a DU may be unnecessary if the corresponding RU remains inactive. This misalignment not only results in inefficient use of resources but also diminishes the potential gains when a UE is admitted to the system. To capture these inter-dependencies, EAPO-NET employs a centralized training phase in which a shared loss couples the two agents' Q-networks, allowing each agent's updates to account for the joint effect of all five action dimensions: RU ON/OFF state, RB blanking ratio, the admission or rejection of a UE requesting access, O-Cloud node ON/OFF state, and DU/CU placement per active RU. This coordination is achieved without inter-agent communication at execution time, preserving the decentralized structure required for deployment as xApps and dApps in a real O-RAN system. Consequently, this approach optimizes UEs' admission for profit, ensuring that each agent's actions enhance operational efficiency while maximizing the potential gains from admitting users to the network.

This work advances the state-of-the-art as follows:

- 1) We formulate the joint RU sleep mode, UE admittance control, and DU/CU VNF placement in a 5G O-RAN system.
- 2) We model the problem as an MDP and design EAPO-NET, a MADRL-based framework deployed as xApps within the Near-RT RIC and as dApps on O-Cloud nodes, enabling real-time optimization under non-stationary network conditions.
- 3) We provide a comprehensive evaluation of EAPO-NET against a B&C optimal solver across solution quality, energy efficiency, and computational scalability, demonstrating that EAPO-NET sustains near-optimal performance while remaining tractable at large-scale deployments where ILP-based approaches become prohibitive.

II. FUNDAMENTALS: PRIMER ON 5G NR, O-RAN AND ENERGY-SAVING IN 3GPP

5G and beyond networks rely on the 5G New Radio (NR) framework, which provides a flexible and scalable frame design capable of supporting a wide range of services, with varying priority levels and distinct QoS requirements [5]. The 5G NR frame structure is organized around a 10 ms radio frame that is partitioned into subframes of 1 ms duration. Each subframe is then divided into slots, where a slot may contain up to 14 symbols [16], [17]. A key feature of this architecture is its adaptable numerology, which defines the granularity of the time-frequency resource grid through different subcarrier spacings. As wireless systems progress toward 6G, it is expected that the existing frame structure

²Co-locating DU/CU virtual functions onto fewer active servers, allowing the remaining servers to be switched off to save energy, e.g., [13], [14].

and numerology concepts will be preserved to maintain compatibility with earlier generations.

O-RAN adopts an open and modular architecture that enables interoperability among multi-vendor equipment while improving network management and performance [3]. Its main components are the RU, DU, and CU. The RU handles lower-layer radio functions, including PHY processing and RF operations at cell sites. The DU processes lower protocol layers, mainly L2 and part of L3, including MAC, RLC, and PDCP functions, while also supporting scheduling, beamforming, and radio resource management [3], [4]. The CU manages upper-layer protocols such as SDAP, RRC, and NAS. O-RAN also integrates the Near-RT RIC, which provides near-real-time control, monitoring, and optimization by interacting with the DU and CU [4]. It hosts xApps for functions such as traffic steering, resource allocation, and energy optimization, while dApps can provide distributed intelligence and decentralized decision-making across the network [4].

Energy saving in 3GPP has become increasingly important due to dense BS deployments and fluctuating traffic demand [18]. A widely studied approach is the ON/OFF sleep mechanism for RUs/BSs, where underutilized nodes enter low-power states during low-traffic periods [10], [11]. These mechanisms may be static or dynamic and often require coordination to avoid coverage gaps or service degradation [10]. In O-RAN, centralized intelligence such as the Near-RT RIC or distributed learning agents can determine RU/BS activation states while considering mobility, QoS, and energy costs. Similarly, O-Cloud energy-saving strategies consolidate DU/CU VNFs onto fewer computing nodes, allowing lightly loaded servers to enter low-power states while preserving service and latency requirements [14], [19], [20]. Complementing node-level energy savings, *RB blanking* operates at the resource-allocation level [9]. Certain RBs are intentionally muted to reduce interference and transmission power, particularly under low-traffic conditions. By selectively blanking RBs, networks can lower power amplifier consumption and inter-cell interference, especially in dense deployments [21]. Combined with sleep modes, RB blanking provides a finer-grained energy-saving mechanism by partially deactivating radio resources without shutting down network elements [8], [9]. Together, these techniques support energy-efficient 5G and beyond networks while maintaining performance.

III. PROBLEM DEFINITION

We first present the system model and its assumptions, and then formulate the problem as a stochastic optimization problem.

A. System Model

The network infrastructure consists of a set $\mathcal{R}$ of RUs, each composed of a set $\mathcal{B}$ of RBs and serving a set $\mathcal{U}$ of UEs over the same geographical area. The computing infrastructure is modeled by a set $\mathcal{V}$ of cloud nodes hosting 5G VNFs, namely DU and CU functions. Each RU must be associated with exactly one DU and one CU, while cloud nodes may host multiple VNFs subject to resource availability. We assume a fully connected topology between RUs and cloud nodes.

Moreover, any network component (RU or cloud node) can be switched to sleep mode, as discussed in Section II. We use variables

$$\mathbf{x} \in \{0, 1\}^{\mathcal{R} \cup \mathcal{V}}, \quad (1)$$

to indicate that a component $i \in \mathcal{R} \cup \mathcal{V}$ is in active mode, i.e., $x_i = 1$, or in sleep mode, i.e., $x_i = 0$.

1) *UE Admission and SLA*: UEs may either be admitted into the system or rejected. Admitted UEs must be associated with exactly one RU, while rejected UEs receive no service. The admission and association decisions are controlled through variables

$$\mathbf{y} \in \{0, 1\}^{R \times U}, \quad (2)$$

where $y_{r,u} = 1$ if UE u is associated with RU r and $y_{r,u} = 0$, otherwise. Hereafter, we refer to $\mathbf{y}$ as the *admission plan*. In 5G systems, SLAs define the QoS guarantees associated with each UE and directly impact the MNO's economic outcomes. In this work, SLA requirements are modeled solely through downlink throughput demands. Therefore, if UE $u \in \mathcal{U}$ is admitted, then

- its *individual profit* $p_u \in \mathbb{R}_{\geq 0}$ is generated for the MNO in line with the SLA, and
- the *number of RBs* $\xi = \{\xi_{r,u} \in \mathbb{N}_{>0} : r \in \mathcal{R}, u \in \mathcal{U}\}$ is consumed from RU r to satisfy its QoS requirements. Parameter $\xi_{r,u}$ changes with the channel conditions³.

2) *RB States*: In this paper, we assume that RUs are equipped with an RB blanking mechanism, as detailed in Section II. When blanking is used, each RB may be (i) available, if it may be scheduled for transmission during the current frame, or (ii) blanked. The available RBs may be in the following states:

- An RB is in *transmitting* state if it is effectively scheduled for data transmission in the current frame. The number of transmitting RBs in RU r is given by

$$g_r^{\text{Tx}}(\mathbf{y}) = \sum_{u \in \mathcal{U}} \xi_{r,u} \cdot y_{r,u}, \quad (3)$$

³We consider that $\xi_{r,u} = \left\lceil \frac{\lambda_u}{\eta_{r,u}} \right\rceil \in \mathbb{N}$ where $\lambda_u \in \mathbb{R}_+$ (in bps) is the average throughput and $\eta_{r,u} \in \mathbb{R}_{\geq 0}$ (in bps/RB) is the achievable wireless link capacity per RB between UE u and RU r .

which depends on the admission plan $\mathbf{y}$.

- (ii) An RB is in an idle state if it is available but not transmitting. The number of idle RBs in RU r is

$$g_r^{\text{Id}}(\mathbf{y}) = g_r^{\text{Av}}(\mathbf{y}) - g_r^{\text{Tx}}(\mathbf{y}), \quad (4)$$

where

$$g_r^{\text{Av}}(\mathbf{y}) = \begin{cases} g_r^{\text{Tx}}(\mathbf{y}), & \text{if blanking is adopted,} \\ B, & \text{otherwise.} \end{cases} \quad (5)$$

In other words, if blanking is adopted, then $g_r^{\text{Id}}(\mathbf{y}) = 0$, i.e., there are no idle RBs.

3) *VNF Placement*: Each cloud node can implement multiple VNFs, and we introduce DU/CU VNF placement variables

$$\mathbf{z}^{\text{VNF}} \in \{0, 1\}^{R \times V}, \text{ VNF} \in \{\text{DU}, \text{CU}\}, \quad (6)$$

to decide whether RU r 's VNF is placed on node i . We denote the set of all VNF placements by $\mathbf{z} = (\mathbf{z}^{\text{DU}}, \mathbf{z}^{\text{CU}})$. The placement of DU and CU functions comes at a computational cost for the cloud nodes. Following a similar approach to prior work (e.g., [22]), we assume that the computing power required to run DU and CU tasks is proportional to the number of transmitting RBs in their associated RU. Therefore, the total computational load (in GOPS) performed by node $i \in \mathcal{V}$ is

$$w_i(\mathbf{y}, \mathbf{z}) = \sum_{r \in \mathcal{R}} g_r^{\text{Tx}}(\mathbf{y}) \cdot (\omega^{\text{DU}} \cdot z_{r,i}^{\text{DU}} + \omega^{\text{CU}} \cdot z_{r,i}^{\text{CU}}), \quad (7)$$

where $g_r^{\text{Tx}}(\cdot)$ is defined in (3) and parameters ω^{DU} and ω^{CU} are the volumes of computing workload per RB (in GOPS/RB) required for running DU and CU tasks, respectively.

4) *Energy-Aware Profit Model*: In this paper, we consider the energy expense to run network operations at the RU-level and at the cloud-level. The energy consumed by RU r is

$$e_r^{\text{RU}}(\mathbf{x}, \mathbf{y}) = [\epsilon^{\text{Av}} \cdot g_r^{\text{Av}}(\mathbf{y}) + \epsilon^{\text{Tx}} \cdot g_r^{\text{Tx}}(\mathbf{y})] \cdot x_r, \quad (8)$$

where ϵ^{Av} and ϵ^{Tx} (both in kWh/RB) are the power consumed to keep one RB in the available state and in the transmitting state [21]. The energy consumed by cloud node i is

$$e_i^{\text{Cloud}}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = (\epsilon_i^{\text{C}} \cdot w_i(\mathbf{y}, \mathbf{z}) + \epsilon_i^{\text{Base}}) \cdot x_i, \quad (9)$$

where ϵ_i^{C} denotes the energy consumed to perform a unit of computational tasks (in kWh/GOPS) and ϵ_i^{Base} represents the baseline power required to keep a cloud node active. The total *energy expense* to run network operations is

$$E(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \sum_{r \in \mathcal{R}} e_r^{\text{RU}}(\mathbf{x}, \mathbf{y}) + \sum_{i \in \mathcal{V}} e_i^{\text{Cloud}}(\mathbf{x}, \mathbf{y}, \mathbf{z}), \quad (10)$$

where $e_r^{\text{RU}}(\cdot)$ and $e_i^{\text{Cloud}}(\cdot)$ are introduced in (8) and (9), respectively.

The *revenue* obtained from admitting UEs into the network corresponds to the total income generated by the admitted UEs, i.e.,

$$H(\mathbf{y}) = \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} p_u \cdot y_{r,u}, \quad (11)$$

where p_u denotes the revenue associated with admitting UE u , which may vary across UEs according to business policies or service priorities.

B. Problem Formulation

Our goal is to maximize the system profit, defined as the difference between the revenue and energy consumption, i.e.,

$$P(\mathbf{x}, \mathbf{y}, \mathbf{z}) = H(\mathbf{y}) - \alpha \cdot E(\mathbf{x}, \mathbf{y}, \mathbf{z}) \quad (12)$$

where $H(\cdot)$ is given by (11), $E(\cdot)$ is given by (10), and α the energy-cost weight (in \$/kWh).

Intuitively, admitting more UEs increases the MNO's revenue, as each admitted UE generates a profit depending on its service characteristics and associated slice. However, serving these UEs requires activating radio and computing resources, thereby increasing the network energy consumption. There is a trade-off: the revenue brought by a UE justifies the additional operational cost required to serve it.

The maximization of (12) is subject to several constraints, which are described next.

C1: At any time, each active RU must be associated with exactly one VNF of each type (i.e., one DU and one CU) deployed on cloud nodes, i.e.,

$$\sum_{i \in \mathcal{V}} z_{r,i}^v = x_r, \forall r \in \mathcal{R}, v \in \{\text{DU}, \text{CU}\}. \quad (13)$$

C2: Any admitted UE can only be associated with an active RU and to one RU only, i.e.,

$$y_{r,u} \leq x_r, \forall r \in \mathcal{R}, \forall u \in \mathcal{U}. \quad (14)$$

$$\sum_{r \in \mathcal{R}} y_{r,u} \leq 1, \forall u \in \mathcal{U}. \quad (15)$$

C3: The set of admitted UEs in each RU is bounded by the existing number of RBs, i.e.,

$$g_r^{\text{Tx}}(\mathbf{y}) \leq B, \forall r \in \mathcal{R}, \quad (16)$$

where $g_r^{\text{Tx}}(\cdot)$ is introduced in (3).

C4: The number of VNFs hosted by cloud node i is limited by its processing capacity μ_i , i.e.,

$$w_i(\mathbf{y}, \mathbf{z}) \leq \mu_i, \quad \forall i \in \mathcal{V}, \quad (17)$$

where we introduce function $w_i(\cdot)$ in (7).

Problem 1. The problem is to maximize the total profit (12) while satisfying system constraints C1–C4.

$$\begin{aligned} & \underset{\mathbf{x}, \mathbf{y}, \mathbf{z}}{\text{maximize}} && P(\mathbf{x}, \mathbf{y}, \mathbf{z}) \\ & \text{subject to} && (1), (2), (6), (13) - (17) \end{aligned}$$

Proposition 1. Problem 1 is $\mathcal{NP}$ -Hard.

We present a proof of Proposition 1 in the Appendix.

In this problem, we have three coupled decisions: (i) network component activation, (ii) UE admission control, and (iii) placement of DU and CU VNFs across the cloud infrastructure. By coordinating these decisions, the system seeks to *consolidate* traffic on a limited subset of infrastructure resources whenever possible, allowing unused RUs or cloud nodes to remain in low-power states. However, Problem 1 faces two major challenges:

- The presence of ξ in the system dynamics makes it difficult to derive deterministic policies that remain effective across all possible realizations.
- In practice, the different decision types are handled by distinct system components that cannot necessarily communicate and coordinate their actions toward a globally optimal solution.

Multi-Agent Reinforcement Learning (MARL) provides a natural framework to address these challenges. First, by learning through repeated interactions with the environment, MARL agents can implicitly adapt to the stochastic dynamics induced by ξ and optimize the expected long-term objective without requiring explicit analytical models. Second, by assigning an autonomous agent to each decision-making component, MARL naturally supports decentralized execution, allowing agents to learn coordinated behaviors while operating with limited or no real-time communication.

IV. EAPO-NET

In this section, we introduce Adaptive Energy-Aware Profit Optimization in Next-Generation RANs (EAPO-NET), a novel MARL framework for tackling Problem 1. The idea is to decompose the global optimization problem into a set of coordinated local decision processes, each handled by a different agent. Each agent performs localized decisions based on partial system observations, while collectively learning a policy that approximates a globally efficient solution. We first explain EAPO-NET's components, then introduce the underlying Markov Decision Processes (MDPs). Finally, we describe the multi-agent algorithm, its properties, and its distribution across the different components.

A. EAPO-NET Components and MDP Formulation

EAPO-NET is a framework where the multiple agents are distributed among O-RAN's (i) Near-RT RIC, in the form of xApps, and (ii) O-Cloud nodes, in the form of dApps. In the following, we define the different types of agents and describe how they relate to Problem 1.

1) *Energy Control (EC)*: Each EC agent is deployed as an xApp that controls a single RU's "sleep mode" and "blanking." We introduce the state space of RU r 's EC agent as $\mathcal{A}_r^{\text{EC}} = [x_r, g_r, \beta_r, \mathbf{y}_r]$, where:

- $x_r \in \{0, 1\}$ indicates if the RU is in active mode,

- $g_r \in \mathbb{Z}_{\geq 0}$ is the amount of transmitting RBs,
- $\beta_r \in [0, 1]$ is the fraction of blanked RBs, and
- $\mathbf{y}_r \subseteq \mathcal{U}$ is the list of UEs associated with RU r .

Moreover, we introduce the action space of RU r 's EC agent as $\mathcal{A}_r^{\text{EC}} = [a_r, b_r]$, where:

- $a_r \in \{\text{ON}, \text{OFF}\}$ denotes the activation action, such that $a_r = \text{ON}$ sets $x_r = 1$ and vice-versa.
- $b_r \in \{0, 0.25, 0.5, 0.75, 1\}$ denotes the blanking action drawn from a discrete set of percentages.

The Lagrangian G^{EC} coming from the EC part is:

$$G^{\text{EC}}(\mathbf{x}, \mathbf{y}) = \sum_{r \in \mathcal{R}} e_r^{\text{RU}}(\mathbf{x}, \mathbf{y}) + \sum_{r \in \mathcal{R}} \alpha_r^{\text{C3}} [g_r^{\text{Tx}}(\mathbf{y}) - B] \quad (18)$$

2) *VNF Control (VC)*: Each VC agent is deployed as a dApp in a given O-Cloud node that controls which VNFs are locally deployed and which ones should be delegated to other O-Cloud nodes. We introduce the state space of node i 's VC agent as $\mathcal{O}_i^{\text{VC}} = [x_i, z_i, w_i]$, where:

- $x_i \in \{0, 1\}$ indicates if the node is in active or sleep mode,
- $z_i \in \{0, 1\}$ indicates which VNFs are being locally hosted, and
- $w_i \in \mathbb{R}_{\geq 0}$ is the amount of resources consumed, analogous to (7).

Moreover, we introduce the action space of node i 's VC agent as $\mathcal{A}_i^{\text{VC}} = [a_i, \mathbf{m}_i^{\text{VNF}}]$, where:

- $a_i \in \{\text{ON}, \text{OFF}\}$ denotes the activation action, such that $a_i = \text{ON}$ sets $x_i = 1$ and vice-versa.
- $\mathbf{m}_i^{\text{VNF}} \in \{0, 1\}^{R \times V}$ denotes the migration action for each VNF. Here, such that $m_{i,r,j}^{\text{VNF}} = 1$ indicates that node i is moving RU r 's VNF to node j .

The Lagrangian G^{VC} coming from the VC part is:

$$G^{\text{VC}}(\mathbf{y}, \mathbf{z}) = \sum_{i \in \mathcal{V}} e_i^{\text{Cloud}}(\mathbf{y}, \mathbf{z}) + \sum_{i \in \mathcal{V}} \alpha_i^{\text{C4}} [w_i(\mathbf{y}, \mathbf{z}) - \mu_i] \quad (19)$$

3) *Admission Control (AC)*: The AC agent is deployed as an xApp that controls which UEs are admitted to the system and which RU they are associated with. We introduce the state space of the AC agent as $\mathcal{O}^{\text{AC}} = [\{\mathcal{O}_r^{\text{EC}} : r \in \mathcal{R}\}, \{\mathcal{O}_i^{\text{VC}} : i \in \mathcal{V}\}, \xi]$, which considers the states of the other agents and the vector of RBs $\xi = \{\xi_{r,u} : r \in \mathcal{R}, u \in \mathcal{U}\}$ required by the UEs. Moreover, we introduce the action space of the AC agent as $\mathcal{A}^{\text{AC}} = \{0, 1\}^{\mathcal{U} \times \mathcal{R}}$, such that each action is a different UE-to-RU association mapping.

The Lagrangian G^{AC} coming from the AC part is:

$$G^{\text{AC}}(\mathbf{x}, \mathbf{y}) = \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} p_u \cdot y_{r,u} + \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} \alpha_r^{\text{C2}} [y_{r,u} - x_r] + \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} \alpha_r^{\text{C2}'} [y_{r,u} - 1] \quad (20)$$

Hence, the total Lagrangian of the system is given by:

$$\begin{aligned}
G(\mathbf{x}, \mathbf{y}, \mathbf{z}) &= G^{\text{EC}} + G^{\text{VC}} + G^{\text{AC}} \\
&= \sum_{r \in \mathcal{R}} e_r^{\text{RU}}(\mathbf{x}, \mathbf{y}) + \sum_{i \in \mathcal{V}} e_i^{\text{Cloud}}(\mathbf{y}, \mathbf{z}) + \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} p_u \cdot y_{r,u} \\
&+ \sum_{r \in \mathcal{R}} \alpha_r^{\text{C3}} [g_r^{\text{Tx}}(\mathbf{y}) - B] + \sum_{i \in \mathcal{V}} \alpha_i^{\text{C4}} [w_i(\mathbf{y}, \mathbf{z}) - \mu_i] \\
&+ \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} \alpha_r^{\text{C2}} [y_{r,u} - x_r] + \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} \alpha_r^{\text{C2}'} [y_{r,u} - 1] \\
&= P(\mathbf{x}, \mathbf{y}, \mathbf{z}) + \sum_{r \in \mathcal{R}} \alpha_r^{\text{C3}} [g_r^{\text{Tx}}(\mathbf{y}) - B] \\
&+ \sum_{i \in \mathcal{V}} \alpha_i^{\text{C4}} [w_i(\mathbf{y}, \mathbf{z}) - \mu_i] + \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} \alpha_r^{\text{C2}} [y_{r,u} - x_r] \\
&+ \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} \alpha_r^{\text{C2}'} [y_{r,u} - 1]
\end{aligned} \tag{21}$$

The MDPs' transition dynamics are unknown and cannot be accurately modeled, making it infeasible to derive or exploit transition probabilities. Consequently, model-based solution methods are not applicable, and the problem must be addressed using model-free learning techniques. Furthermore, the MDPs exhibit very large state-action spaces and significant combinatorial complexity, as discussed in detail in our supplementary report [?]. These characteristics render conventional RL approaches impractical and motivate the adoption of DRL methods.

B. EAPO-NET's Algorithm Description

In a multi-agent setting, a fundamental challenge arises from the non-stationarity of the environment: as each agent updates its policy, the environment appears to change from the perspective of every other agent, breaking the Markov assumption and destabilizing independent learning. Furthermore, while inter-agent coordination is desirable, real-time synchronization at every decision step is impractical given the heterogeneous control loop latencies across agents.

To address these challenges, we draw inspiration from the Value Decomposition Networks (VDN) in the CTDE method [15], which resolves the coordination problem by performing (i) *centralized training*, where the agents' policies are obtained by training their underlying DQNs using a global reward signal and (ii) *decentralized execution*, where the learned policy is applied, requiring no inter-agent message exchange during operation.

Centralized Training Phase: A centralized trainer has access to the joint observation tuple $\langle o_1^t, \dots, o_k^t \rangle$ and maintains one independent utility network $Q_k(o_k, a_k; \theta_k)$ per agent. All agents share a global replay buffer $\mathcal{D}$ that stores transitions $(\mathcal{O}_k(t), \mathcal{A}_k(t), R, \mathcal{O}_k(t+1))$ collected across all k agents. We define the global reward as

$$R_t = G(\mathbf{a}_t(\mathbf{s}_t)) - G(\mathbf{s}_t), \tag{22}$$

where $\mathbf{s}_t \in \mathcal{O}$ is the state at time t , $\mathbf{s}_{t+1} = \mathbf{a}_t(\mathbf{s}_t)$ is the state at time $t+1$ after applying action $\mathbf{a}_t$, and $G(\cdot)$ is the Lagrangian function of Problem 1.

The global reward R_t serves as a common training signal that aligns all agents toward the system-wide objective. Despite sharing the same reward, each agent k learns an independent action-value function by minimizing its own temporal-difference (TD) loss. The target Q-value and loss function are defined as: $y_k = R_t + \gamma \max_{a'_k} Q_k(o_k(t+1), a'_k; \theta_k^-)$

$$\text{and } \mathcal{L}_k(\theta_k) = \mathbb{E}_{\mathcal{D}} \left[(y_k - Q_k(o_k(t), a_k(t); \theta_k))^2 \right].$$

where θ_k^- denotes the parameters of the periodically synchronized target Q-value and γ is the discount factor.

Decentralized Execution Phase: After centralized training, each agent k receives only its trained network θ_k and operates in a fully decentralized manner. At each time slot t , agent k observes solely its local observation $o_k(t)$ and selects its action greedily, without requiring any knowledge of other agents' observations or actions. No inter-agent communication takes place at decision time, and the replay buffer and centralized loss are no longer involved. The IGM condition guarantees that this greedy per-agent maximization remains consistent with the joint optimum found during centralized training.

At execution time, although potential conflicts may arise between agents, they are structurally prevented rather than statistically avoided: the causal ordering enforced within each decision step constrains each agent to act only within the feasible space left by preceding agents, making inter-agent conflicts impossible by construction without requiring any explicit communication.

V. PERFORMANCE EVALUATION

A. Experimental Setup

To evaluate the EAPO-NET framework, we developed a Python-based O-RAN simulation environment instantiated in two configurations: a *small setup* targeting a dense urban micro-cell scenario and a *big setup* representing a larger-scale deployment, with all other system parameters held constant across both configurations.

To implement the MA-DQN learning framework of EAPO-NET, we used PyTorch [23] for scalable, efficient neural network training, enabling practical policy learning. The benchmark optimal solution is obtained by implementing the corresponding ILP described in Problem 1 in Pyomo and solving it with the GLPK solver [24], which relies on a B&C algorithm, to explore the feasible space and guarantee global optimality systematically. Various system parameters are presented in Table I.

B. Convergence Behavior

To assess the convergence behavior of our EAPO-NET framework, we examine three essential metrics

Setup-Dependent Parameters		
Parameter	Small setup	Big setup
Number of RUs $ \mathcal{R} $	4	15
UE population $ \mathcal{U} $ [min, max]	[10, 65]	[50, 150]
UE data rate λ_u [min, max] Mbps	[1.5, 8.25]	[6, 12]

Fixed System Parameters	
Parameter	Value
Number of O-Cloud nodes $ \mathcal{V} $	7
RU capacity B	100 (RBs)
Achievable wireless capacity $\eta_{u,r}$	0.25 (Mbps/RB)
RU Tx energy ϵ^{Tx}	6 (Wh/RB)
O-Cloud Energy $\epsilon_i^{\text{C}}, \epsilon_i^{\text{Base}}$	100, 30 (Wh)
O-Cloud node computing capacity μ_i^{Comp}	300 (GOPS)

Learning Parameters	
Parameter	Value
Epoch	10000
Activation function	ReLU
Optimizer	Adam
Learning rate	$lr = 1 \times 10^{-4}$
$\epsilon_0, \epsilon_{\min}, \epsilon_{\text{decay}}$	1.0, 0.01, 0.995
Discount factor	0.91
Replay buffer, Batch size	$10^6, 128$

TABLE I: Simulation Parameters

within a small setup involving 10 UEs over the training horizon: (i) the average episode reward, (ii) the neural network’s average training loss, and (iii) the evolution of energy consumption during training. These results are reported in Fig. 1(a), Fig. 1(b), and Fig. 1(c), respectively, for both the RB-blanking and non-blanking configurations.

In Fig. 1(a), across problem variants, agents require approximately 2,000 episodes to converge within a 10,000-episode run. The conservative nature of Q-learning and the DNN’s function-approximation capability in DQN lead to stable reward curves with fewer fluctuations but longer convergence times. The loss curves in Fig. 1(b) confirm that the neural network effectively captures system dynamics, as indicated by the consistent decrease in training loss.

Fig. 1(c) illustrates the convergence performance of energy consumption. Both curves typically converge after 2,000 episodes across the problem variants, with the blanking variant showing a clear reduction in energy consumption during training. This reflects the agents’ ability to internalize energy-aware objectives.

Takeaway 1: EAPO-NET can adapt to different operational modes of systems, including exploration of blanking.

C. Analysis of Energy Consumption

1) *Energy consumption as a function of load rate:* We analyze the impact of network load on energy consumption while fixing the average number of UEs to 10. The evaluation considers: (i) the received load per UE (bps), (ii) the energy consumption per bps, and (iii) the per-slot decision time of each approach. Results for

both RB-blanking and non-blanking configurations are shown in Fig. 2.

Fig. 2(a) shows that total energy consumption increases with the load rate due to higher traffic demand and RB utilization. Curve intersections at 2.5 and 5 Mbps correspond to RU saturation points, where a single RU exhausts its RB capacity. Beyond saturation, blanking cannot yield additional savings, and the two curves overlap. When extra RUs are activated, the load is redistributed, creating new opportunities for blanking until the next saturation point. Under high traffic (6.5–8.25 Mbps), blanking achieves energy savings of 10 – 20% with EAPO-NET, closely matching the B&C solution (11 – 20%).

Fig. 2(b) shows that energy consumption per bps decreases as the load rate increases, indicating improved RB utilization and efficiency. Fig. 2(c) reports the per-slot decision time. EAPO-NET requires only 1 – 7 ms with blanking and 1 – 5 ms without blanking, whereas B&C requires 0.35 – 165 s and 0.17 – 149 s, respectively. The additional blanking decisions have a limited impact on EAPO-NET but significantly increase B&C’s optimization time.

Takeaway 2: EAPO-NET achieves near-optimal performance, with an optimality gap of 0.7 – 3% (blanking) and 0.8 – 2.2% (non-blanking), while providing near-instantaneous decisions. In contrast, B&C attains optimal solutions but incurs substantially higher computation times, which grow with problem size. Increasing the load rate has only a limited impact on decision latency.

2) *Energy consumption as a function of UEs density:* For a fixed average load per UE of 1.5 Mbps, we evaluate the energy consumption of EAPO-NET framework, based on (i) UEs density, (ii) consumption per UE, and (iii) the per-slot decision time of each approach. Results for both RB-blanking and non-blanking configurations are shown in Fig. 3.

Fig. 3(a) illustrates that energy consumption grows with the number of UEs due to increased processing and RB utilization. The curves intersect at UE loads of 15, 30, 48, and 65, which correspond to RU saturation points: once an RU is fully utilized, blanking cannot further reduce energy, as savings mainly arise during underloaded periods. As load increases, additional RUs are activated, creating new operating regions where blanking again provides energy gains until the next saturation point. Notably, in the 33–48 UE range, blanking attains peak energy savings of 18 – 4% with EAPO-NET and 18 – 3% with the optimal model. Fig. 3(b), we analyze the energy consumption per UE to address the question: *How efficiently does the system serve each UE as load increases?* The figure shows that energy per UE decreases as the number of UEs increases, driven by improved RBs utilization and amortization of static energy costs (for active RU). In Fig. 3(c), we explore

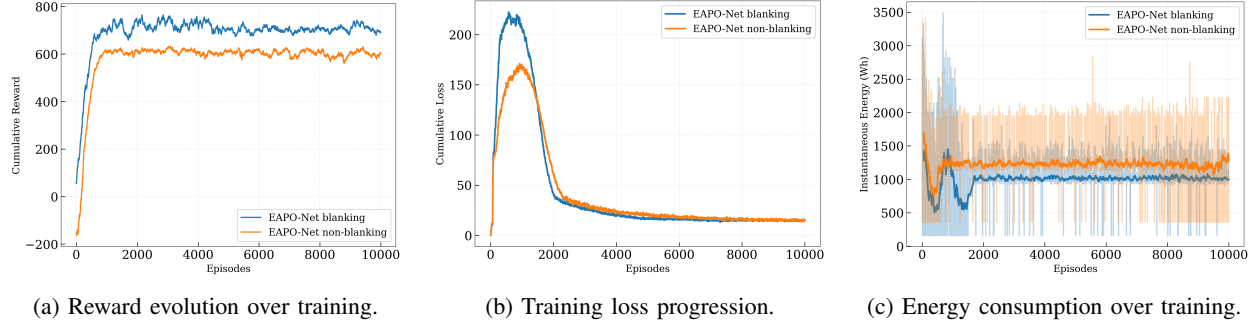


Fig. 1: Analysis of reward, loss, and energy consumption.

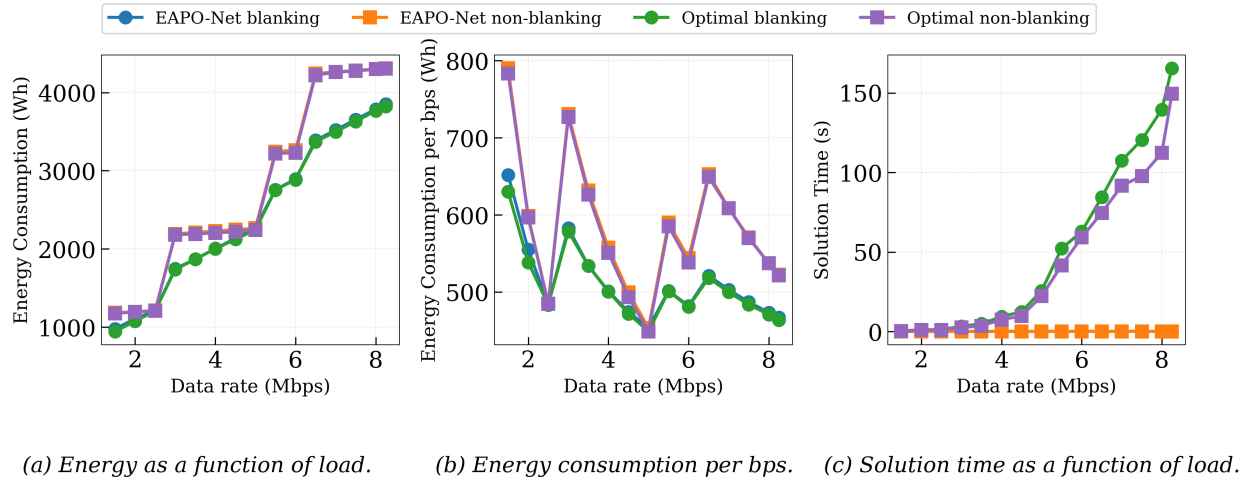


Fig. 2: Energy consumption under various network loads.

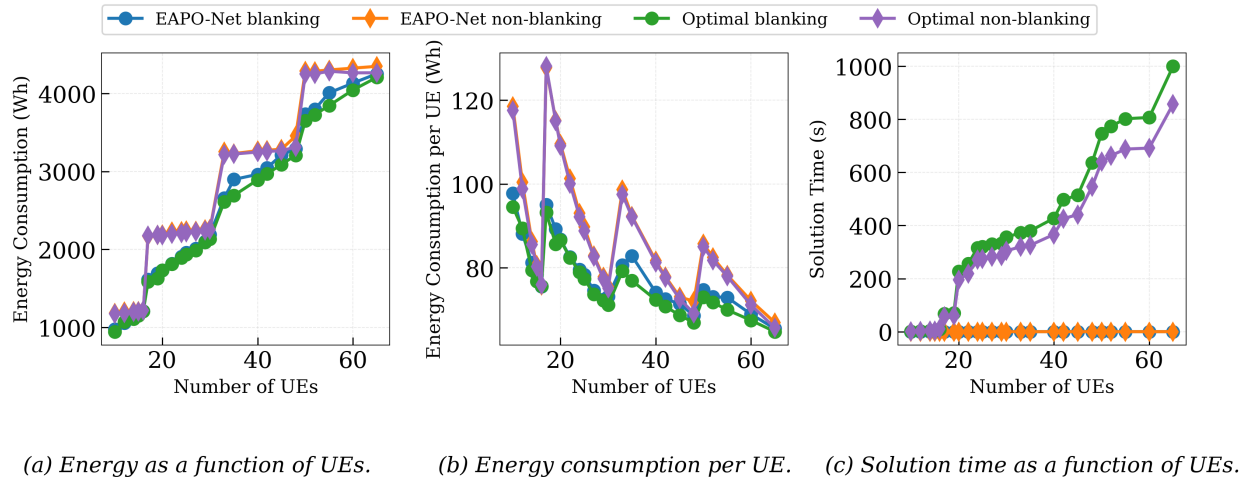


Fig. 3: Energy consumption under various UE sets.

the time taken at every time slot as the number of UEs increases. It can be shown that for the blanking variant, EAPO-NET achieves latencies of $1.9 - 13\text{ ms}$ compared to $0.35 - 999\text{ s}$ for the B&C. Under the non-blanking variant, EAPO-NET achieves latencies of $1.7 - 12.32\text{ ms}$ compared to $0.3 - 856\text{ s}$ for B&C. The advantage of using EAPO-NET over B&C becomes more evident as we increase the problem's size (number of UEs); the problem's time complexity increases really fast with the problem's size, and it is harder for B&C to catch up (too many branches to explore).

Takeaway 3: EAPO-NET not only achieves close optimal performance but also demonstrates a significant speed advantage, highlighting its responsiveness to dynamic conditions.

D. Robustness and Scalability Evaluation

To evaluate the robustness and scalability of EAPO-NET, we conduct experiments within a big setup. We evaluate its behavior under increasing system complexity along two orthogonal dimensions. Our exploration centers around the following key questions:

- (i) Does EAPO-NET maintain robust and feasible decisions under large-scale deployments?
- (ii) How does the solution quality of EAPO-NET scale with network size compared to the B&C optimum?
- (iii) What is the computational scalability of EAPO-NET relative to the B&C optimum?

Network Load-size scalability. We fix the average number of UEs to 20 and vary the average per-UE load from 6 to 12 Mbps in steps of 0.5 Mbps. Since each UE requires $\text{load}/0.25$ RBs to satisfy its demand, this corresponds to an RB demand ranging from 10 to 48 RBs per UE, progressively tightening the resource budget available for RB blanking and Cloud nodes consolidation.

Solution quality. Fig. 4(a) reports the total network energy consumption as a function of network load. Across all the loads, EAPO-NET closely tracks the B&C optimum, achieving an optimality gap below $0.5 - 0.1\%$ for problem variant blanking and $0.4 - 1.25\%$ for non-blanking variant. As demand increases beyond 10.5 Mbps per UE, the gap between EAPO-NET and the B&C widens moderately, reflecting the increased difficulty of coordinating blanking decisions under tighter RB constraints. Nevertheless, EAPO-NET achieves close performance across the entire load range, confirming that energy savings remain achievable even under high traffic pressure. The energy efficiency metric expressed in Wh per bit further highlights this trend. As shown in Fig. 4(b), both EAPO-NET and the B&C exhibit an improvement in energy per bit as load increases, consistent with the amortization of RU static power costs over a larger traffic volume. As load increases and the blanking margin shrinks, this gap between blanking and non-blanking narrows, reflecting the reduced opportunity

to switch off RUs under tight RB demand. Throughout the load range, EAPO-NET blanking variant closely tracks the B&C blanking variant.

Computational scalability. Regarding decision latency, Fig. 4(c), the per-slot decision time of EAPO-NET remains relatively constant across all load levels, as the neural network's forward pass is unaffected by traffic intensity. In contrast, the solve time for B&C increases significantly with load due to tighter feasibility constraints, which necessitate exploring a larger portion of the solution space before convergence. At a load of 12 Mbps per UE, the average time required is 999 seconds per time slot, which is not suitable for real-time O-RAN control loops.

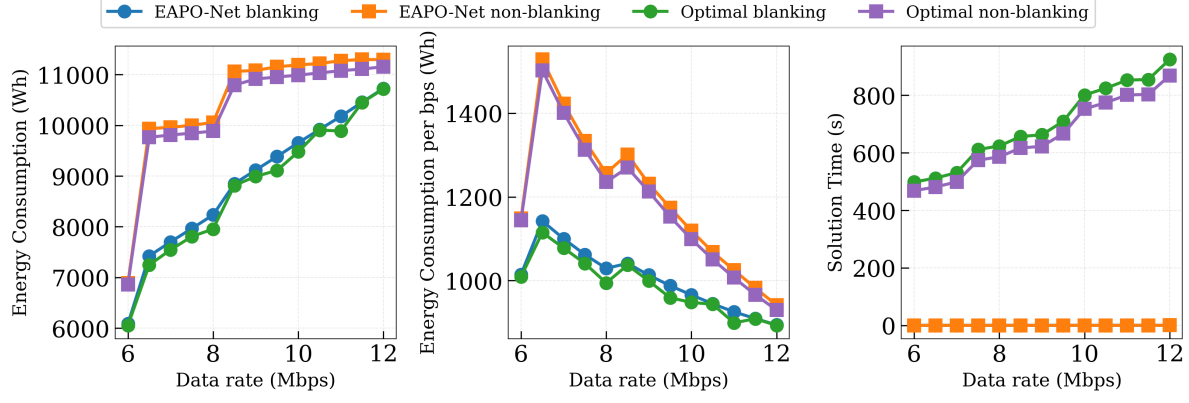
Takeaway 4: Under high network load and across all evaluated problem variants, EAPO-NET achieves near-optimal solutions with a significantly lower time step than B&C, confirming its practical viability for real 5G traffic conditions.

Problem-size scalability. To further validate the scalability of EAPO-NET, we conduct a comparative analysis in which the average per-UE load demand is fixed at 2.5 Mbps, corresponding to 8 RBs per UE, while the average number of UEs varies from 50 to 150 in steps of 10. This setup isolates the effect of problem size from load intensity, as each additional UE uniformly expands the joint UE-RU association, VNF placement, and RB blanking decision space without altering the per-UE resource footprint.

Solution quality. Fig. 5(a) and Fig. 5(b) reports the energy consumption, total and per-UE for EAPO-NET and B&C across all network sizes. At small scales, EAPO-NET with blanking variant closely tracks B&C with variant blanking within, while both blanking variants consistently outperform their non-blanking counterparts. As the number of UEs grows, the optimality gap remains bounded within $2.1 - 2.2\%$, demonstrating that EAPO-NET solution quality does not degrade as the problem scales. This confirms that the learned policy generalizes to configurations beyond those encountered during training without meaningful loss in energy performance.

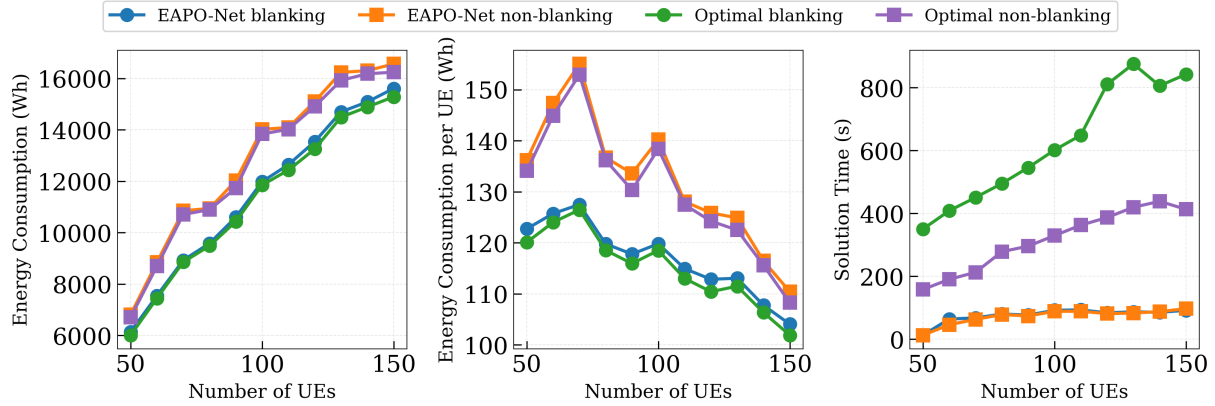
Computational scalability. Fig. 5(c) reports the per-slot decision time of both approaches. B&C solve time grows rapidly with the number of UEs, reaching seconds per slot at 150 UEs, driven by the combinatorial expansion of the search space. In contrast, EAPO-NET per slot time remains approximately constant at $35 - 40\text{ ms}$ across the entire range, as the neural network forward pass is agnostic to problem size. At 150 UEs, EAPO-NET is $\times 4.10^3$ faster than B&C, a gap that widens consistently with network size, making B&C increasingly unsuitable for real-time control at large deployments.

Takeaway 5: As the number of UEs grows, B&C solve time increases sharply. In contrast, EAPO-NET



(a) Energy as a function of load. (b) Energy consumption per bps. (c) Solution time as a function of load.

Fig. 4: Network-load size scalability evaluation.



(a) Energy as a function of UEs. (b) Energy consumption per UE. (c) Solution time as a function of UEs.

Fig. 5: Problem-size scalability evaluation.

per-slot time remains constant, yet EAPO-NET sustains a bounded optimality gap throughout, confirming its viability for large-scale deployments where B&C becomes computationally prohibitive.

E. Sensitivity Analysis

We now evaluate distinct features of EAPO-NET.

Fig. 6(a) illustrates the RB utilization breakdown over training episodes. During the early exploration phase (episodes 0–500), high variance across all curves reflects the agent operating with multiple active RUs before learning to consolidate traffic. Beyond episode 500, the policy stabilizes: the Tx curve converges to approximately 60–70 RBs per episode, consistent with a single active RU serving the admitted UE population at 60–70% load, while the blanked curve rises to match the complementary idle fraction, and the idle (wasted) curve collapses to near zero. This convergence pattern confirms

that the RU agent has jointly learned two complementary behaviors: consolidating all traffic onto a single RU to sleep the remaining three, and blanking nearly all idle RBs on the active RU to eliminate residual radio-frequency energy waste.

Fig. 6(b) shows the per-UE admission rate as a function of α described in Eq. (12). Three key observations emerge. First, the agent learns the correct UE-priority ordering, shedding *priority 3* traffic first, followed by *priority 2*, while protecting *priority 1*, purely from the reward signal, without any hardcoded priority rule, validating that the Lagrangian formulation correctly encodes UE value into the Q-learning objective. Second, the admission curves exhibit sharp inflection points precisely at the analytical break-even thresholds $\alpha_{priority3}^* = 1.94$, $\alpha_{priority2}^* = 2.78$, and $\alpha_{priority1}^* = 4.17$ \$/Wh, confirming that the learned policy is consistent with the condition $\alpha^* = H(y)/E(x, y, z)$ derived from the

Eq. (12). Third, even at $\alpha = 4.78$ where all three priorities exceed their break-even, *priority 1* admission stabilizes at 60% rather than collapsing, demonstrating that the agent balances energy savings against revenue loss rather than maximizing either objective in isolation, and confirming that α provides a tunable knob to slide the operating point along the energy-revenue Pareto frontier.

Takeaway 6: A weighted reward function with an energy-cost factor α can naturally drive UE-aware admission and energy-efficient behavior without predefined rules. However, α must be properly tuned according to the analytical threshold α^* ; otherwise, the agent may either over-admit traffic or become overly restrictive, reducing revenue.

VI. RELATED WORK

Mobile networks face a critical challenge today: **Energy consumption** is rapidly increasing. At the same time, traffic becomes more dynamic and unpredictable. Current O-RAN control strategies are mostly *static or rule-based*, which makes them too rigid. As a result, they often keep RAN and computing resources active even when traffic is low, wasting energy. Existing efforts to improve RAN energy efficiency rely merely on cell sleeping (Cell Switching ON/OFF) and workload consolidation.

A large body of work focuses on exploiting traffic fluctuations to deactivate base stations or sectors. For instance, [8], [25] jointly optimize cell clustering and BS sleeping to consolidate load during medium- and long-term low-traffic periods. However, such consolidation generally requires strong inter-cell coordination, which is often unrealistic in dense or heterogeneous deployments.

At shorter timescales, fine-grained opportunities for EE arise without full cell shutdown. Cell Discontinuous Transmission (DTX) [6], [7] enables sub-TTI micro-sleep at the RU during idle periods. Building on this, several O-RAN works [26]–[29] design xApps for real-time Advanced Sleep Modes (ASMs) and energy-aware scheduling. However, most of these approaches assume idealized, perfect cell overlapping or near-perfect UE handover flexibility, which is often infeasible in realistic deployments. When capacity consolidation fails (e.g., due to coverage or QoS constraints), these techniques offer limited benefit. The study [30], for instance, optimizes UE–RU association and RB allocation to reduce fronthaul load and improve network utilization. More advanced works [31]–[33] jointly optimize RB allocation, MCS selection, and multi-RU beamforming to maximize proportional fairness. These approaches excel at QoS-driven RRM, but rarely address fine-grained energy-saving mechanisms such as RB-level adaptation. A parallel line of work addresses the energy footprint of O-Cloud infrastructure. OrchestRAN [19] provides orchestration tools for deploying latency-aware inference

pipelines with flexible timing constraints. Works such as [14], [20] optimize VNF placement and reconfiguration costs in O-RAN. RB blanking, in contrast, offers an orthogonal method: saving energy within an active RU without requiring cell consolidation or ASM triggering. This micro-level mechanism remains underexplored in the context of O-RAN [34], [35].

Building on these advances, EAPO-NET advances RL-enabled energy-saving in O-RAN through three key contributions: (i) a joint optimization framework for RU sleep control (including ON/OFF and RB blanking), UE admittance control, and DU/CU VNF placement; (ii) a distributed multi-agent architecture that decomposes the problem across three heterogeneous agent classes: RU xApp agents, an admission xApp agent, and VNF dApp agents jointly trained under the CTDE paradigm using VDN and executed independently without inter-agent communication; (iii) a comprehensive evaluation against exact ILP solutions to assess scalability, adaptability, and solution efficiency.

VII. CONCLUSION

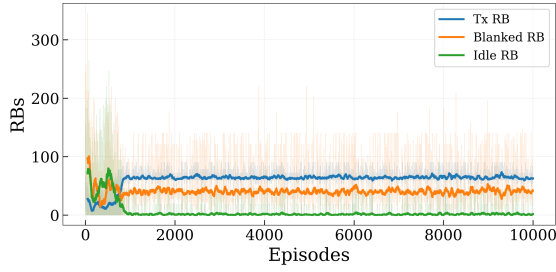
This paper presented EAPO-NET, a joint optimization framework for RU sleep mode, UE admittance control, and DU/CU VNF placement in a 5G O-RAN system.

By decomposing the problem across three heterogeneous agent classes — RU xApp agents, an admission xApp agent, and nodes dApp agents — EAPO-NET addresses the fundamental tension between energy savings, network profit, and SLA compliance that MNOs face under dynamic traffic conditions. Training EAPO-NET under the CTDE paradigm with a VDN global objective ensures that agents coordinate implicitly during training while operating fully independently at deployment, with no inter-agent communication required at decision time.

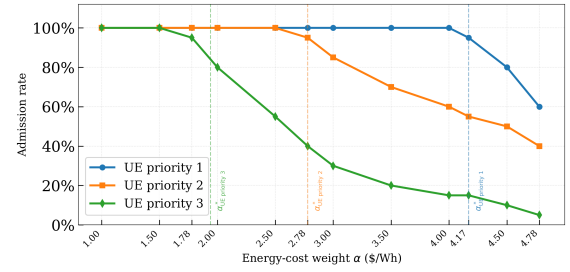
Through rigorous experimentation on O-RAN networks of various sizes, we demonstrated the effectiveness of EAPO-NET to learn near-optimal control policies. Compared with exact ILP solutions, EAPO-NET achieves comparable optimization quality while substantially improving scalability and adaptability to time-varying network conditions. As future work, we plan to extend EAPO-NET to multi-cell deployments where inter-cell interference and handover decisions introduce additional coordination challenges, and to investigate online fine-tuning mechanisms at the non-RT RIC to adapt the frozen policy to long-term distribution shifts without full retraining.

REFERENCES

- [1] S. Han, S. Bian *et al.*, “Energy-efficient 5g for a greener future,” *Nature Electronics*, vol. 3, no. 4, pp. 182–184, 2020.
- [2] T. Li, L. Yu, Y. Ma, and T. e. a. Duan, “Carbon emissions and sustainability of launching 5g mobile networks in china,” *arXiv preprint arXiv:2306.08337*, 2023.
- [3] O.-R. Alliance, “O-ran whitepaper - building the next generation ran,” October 2018, accessed: 2026-05-15. [Online]. Available: <https://www.o-ran.org/resources>



(a) Evolution of Tx RBs, Blanking RBs, and Idle RBs throughout execution.



(b) Admission rate as a function of energy-cost weight α (\$/Wh).

Fig. 6: Per-UE type admission rate as a function of the energy-cost weight trade-off α . Vertical dashed lines mark the break-even thresholds $\alpha^* = H(\mathbf{y})/E(\mathbf{x}, \mathbf{y}, \mathbf{z})$ beyond which rejecting a UE of a certain type becomes energy-profitable. The agent learns to shed low-value traffic *priority 3* first, followed by *priority 2*, while protecting high-revenue *priority 1* users until $\alpha \geq \alpha^*_{\text{priority 1}} = 4.17$ \$/Wh.

- [4] M. Polese, L. Bonati, S. D'oro, S. Basagni, and T. Melodia, "Understanding o-ran: Architecture, interfaces, algorithms, security, and research challenges," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 2, pp. 1376–1411, 2023.
- [5] O.-R. Alliance, "O-ran slicing architecture 14.01," June 2025, technical Specification, O-RAN.WG1.TS.Slicing-Architecture-R004-v14.01, Adopted by ATIS, R004.
- [6] S. Tombaz, S.-w. Han, K. W. Sung, and J. Zander, "Energy efficient network deployment with cell dtx," *IEEE Communications Letters*, vol. 18, no. 6, pp. 977–980, 2014.
- [7] P. Chang and G. Miao, "Optimal operation of base stations with deep sleep and discontinuous transmission," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 11, pp. 11113–11126, 2018.
- [8] J. Kim, H.-W. Lee, and S. Chong, "Traffic-aware energy-saving base station sleeping and clustering in cooperative networks," *IEEE Transactions on Wireless Communications*, vol. 17, no. 2, pp. 1173–1186, 2017.
- [9] V. Kasuluru, L. Blanco, C. J. Vaca-Rubio, and E. Zeydan, "On the impact of prb load uncertainty forecasting for sustainable open ran," in *IEEE PIMRC 2024*, pp. 1–7.
- [10] M. Feng, S. Mao, and T. Jiang, "Boost: Base station on-off switching strategy for energy efficient massive mimo hetnets," in *IEEE INFOCOM 2016*, pp. 1–9.
- [11] P. Shen, Y. Shao, Q. Cao, and L. Lu, "Dynamic gnodeb sleep control for energy-conserving radio access network," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 4, pp. 1371–1385, 2024.
- [12] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "Colo-ran: Developing machine learning-based xapps for open ran closed-loop control on programmable experimental platforms," *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 5787–5800, 2023.
- [13] R. Joda, T. Pamuklu, P. E. Iturria-Rivera, and M. Erol-Kantarci, "Deep reinforcement learning-based joint user association and cu-du placement in o-ran," *IEEE Transactions on Network and Service Management*, vol. 19, no. 4, pp. 4097–4110.
- [14] A. C. Bessala, G. Iecker Ricardo, and G. Jakllari, "E-MATRO: an adaptive and Energy-Efficient framework for slicing within the O-RAN architecture," in *IFIP Networking 2025*.
- [15] P. Sunehag, G. Lever, A. Gruslys, W. M. Czarnecki, V. Zambaldi, M. Jaderberg, M. Lanctot, N. Sonnerat, J. Z. Leibo, K. Tuyls et al., "Value-decomposition networks for cooperative multi-agent learning," *arXiv preprint arXiv:1706.05296*, 2017.
- [16] ETSI, "5g nr: Physical channels and modulation (3gpp ts 38.211 version 16.2.0 release 16)," https://www.etsi.org/deliver/etsi_ts/138200_138299/138211/16.02.00_60/ts_138211v160200p.pdf.
- [17] D. Parruca and J. Gross, "Throughput analysis of proportional fair scheduling for sparse and ultra-dense interference-limited ofdma/lte networks," *IEEE Transactions on Wireless Communications*, vol. 15, no. 10, pp. 6857–6870, 2016.
- [18] O. Alliance, "Potential energy savings features in o-ran," *White Paper*, no. 2025-01, 2025.
- [19] S. D'Oro, L. Bonati, M. Polese, and T. Melodia, "Orchestrator: Network automation through orchestrated intelligence in the open ran," in *IEEE INFOCOM 2022*, pp. 270–279.
- [20] G. Almeida, M. Abdel-Rahman, and K. Cardoso, "Constraint-aware deep reinforcement learning for vran dynamic placement," SBC, 2024, pp. 337–350.
- [21] M. Lauridsen, A. R. Jensen, and P. Mogensen, "Reducing lte uplink transmission energy by allocating resources," in *IEEE VTC 2011*, pp. 1–5.
- [22] R. Singh, C. Hasan, X. Foukas, M. Fiore, M. K. Marina, and Y. Wang, "Energy-efficient orchestration of metro-scale 5g radio access networks," in *IEEE INFOCOM 2021*, 2021, pp. 1–10.
- [23] U. L. Imaz and F. T. Garcia, "Deep reinforcement learning in pytorch," 2019.
- [24] W. E. Hart, J.-P. Watson, and D. L. Woodruff, "Pyomo: modeling and solving mathematical programs in python," *Mathematical Programming Computation*, vol. 3, no. 3, pp. 219–260, 2011.
- [25] F. E. Salem, T. Chahed, E. Altman, A. Gati, and Z. Altman, "Optimal policies of advanced sleep modes for energy-efficient 5g networks," in *IEEE NCA 2019*, pp. 1–7.
- [26] J. X. Salvat Lozano, J. A. Ayala-Romero, A. Garcia-Saavedra, and X. Costa-Perez, "Kairos: Energy-efficient radio unit control for o-ran via advanced sleep modes," in *IEEE INFOCOM 2025*, pp. 1–10.
- [27] A.-K. Dang, H. Khalifé, M. Sintorn, S. Rovedakis, and S. Secci, "Data-driven energy optimization in mobile networks with user experience guarantees," in *IEEE INFOCOM 2025*, pp. 1–10.
- [28] J. A. Ayala-Romero, L. Lo Schiavo, A. Garcia-Saavedra, and X. Costa-Perez, "Mean-field multi-agent contextual bandit for energy-efficient resource allocation in vrans," in *IEEE INFOCOM 2024*, pp. 911–920.
- [29] W. T. Pires-JR, G. M. Almeida, S. L. Corrêa, C. B. Both, L. L. Pinto, and K. V. Cardoso, "Optimizing energy consumption for vran placement in o-ran systems with flexible transport networks," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 4279–4294, 2025.
- [30] B. Han, V. Sciancalepore, D. Feng, X. Costa-Perez, and H. D. Schotten, "A utility-driven multi-queue admission control solution for network slicing," in *IEEE INFOCOM 2019*, pp. 55–63.
- [31] O. Orhan, V. N. Swamy, T. Tetzlaff, M. Nassar, H. Nikopour, and S. Talwar, "Connection management xapp for o-ran ric: A graph neural network and reinforcement learning approach," in *IEEE ICMLA 2021*, pp. 936–941.
- [32] D. Harutyunyan, R. Behraves, and N. Slamnik-K, "Cost-efficient placement and scaling of 5g core network and mec-enabled application vnfs," in *IFIP/IEEE IM 2021*, pp. 241–249.

- [33] M. K. Motalleb, V. Shah-Mansouri, S. Parsaeefard, and O. L. A. López, "Resource allocation in an open ran system using network slicing," *IEEE Transactions on Network and Service Management*, vol. 20, no. 1, pp. 471–485, 2022.
- [34] F. Salahdine, J. Opadere, Q. Liu, T. Han, N. Zhang, and S. Wu, "A survey on sleep mode techniques for ultra-dense networks in 5g and beyond," *Computer Networks*, vol. 201, p. 108567, 2021.
- [35] R. Smith, M. Farshchian, C. Freeberg, T. Machacek, and D. Li, "Enabling technologies for spectrum sharing with oran," in *IEEE DySPAN 2024*, pp. 102–107.